Enhancing Image Processing Efficiency for MRI Data: Comparative Analysis of Threading and Parallel Processing

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***Abstract*—Magnetic Resonance Imaging (MRI) is pivotal in diagnosing brain tumors, offering precise imaging critical for early intervention. However, the computational demands of MRI data processing present significant challenges, particularly for real-time clinical applications. This study investigates a hybrid framework that integrates U-Net-based segmentation with CNN-based classification, enhanced by threading and GPU acceleration to optimize efficiency. By employing parallel processing techniques, the framework achieved a 35% reduction in preprocessing time and a 40% reduction in training time compared to non-threaded implementations.**

The hybrid model demonstrated a classification accuracy of 99.6%, excelling in the detection of glioma, meningioma, pituitary tumors, and non-tumor cases. Segmentation outputs provided critical spatial features, enriching the classification model's predictive capabilities. Despite these advancements, challenges in glioma and pituitary tumor classification were identified, revealing opportunities for further improvement.

This research underscores the potential of parallel processing techniques, including threading and GPU acceleration, in enhancing MRI workflow efficiency. By addressing computational bottlenecks and improving diagnostic precision, the proposed framework lays the groundwork for scalable, real-time medical imaging solutions.

***Keywords*—MRI, Brain Tumor Detection, U-Net, CNN, Hybrid Models, Threading, GPU Acceleration**

I. Introduction

Brain tumors are among the most significant contributors to neurological morbidity and mortality worldwide, accounting for a substantial proportion of cancer-related deaths. Early and accurate detection of brain tumors is crucial for timely intervention and effective treatment. Over the past few decades, Magnetic Resonance Imaging (MRI) has emerged as the gold standard for brain tumor diagnosis, owing to its ability to provide high-resolution, non-invasive, and detailed images of the brain's structure. MRI scans are instrumental in differentiating healthy from diseased tissues, enabling clinicians to identify abnormalities such as tumors with precision [1], [2].

Despite its diagnostic capabilities, the increasing complexity and resolution of MRI data impose significant computational demands. Processing high-resolution MRI scans involves challenges related to computational time and resource utilization. In clinical environments, delayed diagnosis due to slow processing can have critical consequences, particularly in acute care settings where rapid decision-making is vital. Traditional approaches to MRI analysis, including manual interpretation and early machine learning models, have proven to be both time-intensive and prone to variability. Thus, there is a pressing need for efficient systems capable of processing vast amounts of data swiftly and accurately [3], [4].

Artificial Intelligence (AI) and deep learning techniques have emerged as transformative tools in overcoming these challenges. Advanced models, particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable success in feature extraction, tumor segmentation, and classification tasks. However, existing approaches often focus on individual tasks without considering the efficiency of the entire MRI workflow. Additionally, scalability to real-time clinical applications remains an underexplored area.

This research addresses these limitations by investigating the use of threading and parallel processing techniques to enhance the performance of MRI workflows. By leveraging threading, GPU acceleration, and distributed computing through Message Passing Interface (MPI), this study aims to optimize preprocessing, feature extraction, and training processes. Threading enables parallel execution of tasks, efficiently utilizing multi-core processors and GPUs, while MPI facilitates distributed processing of large datasets across multiple machines, reducing computational bottlenecks [5], [6].

This study evaluates the performance of a hybrid framework that integrates U-Net-based segmentation with CNN-based classification, comparing threaded and non-threaded implementations. The research hypothesizes that incorporating threading and parallel processing will not only enhance processing efficiency but also improve diagnostic accuracy, making real-time deployment in clinical settings feasible.

The objectives of this study are to:

1) Assess the impact of threading and parallel processing on the efficiency of MRI workflows, with a focus on preprocessing, feature extraction, and training stages.

2) Compare the performance of hybrid U-Net and CNN models with and without threading to quantify improvements in computational efficiency.
3) Evaluate the scalability and practical applicability of these optimization techniques in real-time clinical environments.
4) Develop a comprehensive framework for integrating threading and parallel processing into MRI-based diagnostic workflows, emphasizing improved resource utilization and diagnostic accuracy.

Through this comparative analysis, this research aims to establish an efficient approach to MRI data processing, paving the way for advancements in brain tumor detection and clinical decision-making.

II. Literature Review

Developments in MRI-Based Brain Tumor Detection

Research on brain tumor detection has evolved significantly over the years, transitioning from classical machine learning approaches to advanced deep learning and hybrid frameworks. This literature review summarizes key contributions in the field, arranged chronologically to illustrate the advancements in methodologies.

2015–2018: Early Machine Learning Techniques

Initial approaches primarily relied on classical machine learning methods with handcrafted feature extraction. Yang et al. (2016) developed a CNN integrated with Local Binary Patterns (LBP) for texture feature extraction, achieving an accuracy of 95.2% [7]. This approach demonstrated the potential of CNNs for tumor classification but lacked support for multiclass datasets and real-time deployment. Similarly, SVM and Random Forest-based models achieved moderate success by leveraging shape and texture features, though their scalability and generalizability were limited.

2018–2020: Emergence of Deep Learning Models

The introduction of deep learning models marked a turning point in MRI-based tumor detection. Amin et al. (2019) proposed a hybrid CNN-K-means clustering approach, achieving 96.73% accuracy, effectively combining segmentation and classification tasks [5]. Zhang et al. (2019) utilized transfer learning for metastases detection, achieving 93.8% accuracy but faced challenges in adapting to diverse datasets [8]. Hybrid architectures, such as the CNN-inception module proposed by Çınar and Yildirim (2019), achieved 92.5% accuracy, although interpretability and optimization were noted as areas for improvement [9].

2020–2022: Advanced Hybrid and Transfer Learning Approaches

The period saw significant advancements in combining hybrid models and transfer learning. Hashemzehi et al. (2020) introduced a hybrid CNN-NADE model, achieving 94.3% accuracy but required validation across diverse datasets [10]. Khairandish et al. (2021) developed a CNN-SVM approach for tumor detection and classification, achieving 94.2% accuracy by utilizing threshold segmentation and SVM-based classification [11]. Similarly, Brindha et al. (2021) emphasized the importance of data augmentation in improving CNN-based classification, though their work lacked explicit performance metrics [4].

Chattopadhyay and Maitra (2022) incorporated dropout regularization and batch normalization into CNN architectures, achieving an accuracy of 97.01% [12]. Mahmud et al. (2022) utilized transfer learning in a hybrid CNN-SVM model, achieving 98.3% accuracy for multiclass tumor classification [3]. Mahjoubi et al. (2022) extended this work by integrating residual layers, achieving 98.7% accuracy, but noted scalability challenges for resource-constrained environments [13].

2023–2024: Lightweight Models and Real-Time Applications

Recent advancements have focused on optimizing models for real-time clinical environments. Almufareh et al. (2023) employed YOLOv5 for tumor segmentation and classification, achieving a mean average precision (mAP) of 98.2% [6]. Xu and Mohammadi (2023) explored lightweight architectures like MobileNetV2 optimized through contracted fox optimization algorithms, achieving 95.6% accuracy [14]. Asiri et al. (2023) introduced a dual-module CNN architecture for image enhancement and classification, achieving 98.5% accuracy while addressing dataset imbalances [15].

Al-Shahrani et al. (2024) emphasized the importance of augmented datasets for binary tumor classification, achieving 97.3% accuracy [16]. Khaliki et al. (2024) conducted a comparative analysis of ResNet50 and lightweight CNNs, achieving 96.4% accuracy while highlighting the need for optimization in resource-constrained environments [2]. Mathivanan et al. (2024) applied transfer learning with ResNet and VGG models, achieving 97.6% accuracy, though their generalizability required further validation [17].

Recent contributions, such as Amin et al. (2024), proposed multimodal feature fusion with SVM classifiers, achieving 94.7% accuracy [18]. Anantharajan et al. (2024) introduced an ensemble CNN-Random Forest approach, achieving 96.5% accuracy but requiring further optimization for edge computing [19].

Summary of Advancements

These developments illustrate the rapid evolution of MRI-based tumor detection methods, focusing on enhancing diagnostic accuracy and computational efficiency. The shift toward hybrid models and lightweight architectures highlights the field's emphasis on scalability and real-time clinical applicability. Building upon these advancements, this study introduces a hybrid framework combining U-Net segmentation and CNN classification, leveraging threading and GPU acceleration to optimize performance. This approach aims to streamline MRI workflows and improve diagnostic outcomes in clinical settings.

III. Methodology

1. Tumor Segmentation Framework

1.1 Selection of Segmentation Model: The U-Net architecture with an EfficientNet-B7 encoder, pre-trained on ImageNet, was employed for tumor segmentation. This encoder-decoder structure is widely recognized for its efficacy in medical image processing, enabling it to capture fine features and produce accurate segmentations. The decoder was tailored for MRI segmentation tasks, ensuring effective tumor detection.

1.2 Segmentation Pipeline: The segmentation process consisted of the following steps:

- **Preprocessing and Normalization:**
  - MRI scans were resized to 256×256 to match the input size expected by the model.
  - Noise reduction, contrast enhancement, and normalization techniques were applied to standardize the dataset.
  - Augmentation techniques, such as rotation, scaling, and flipping, were implemented to enhance variability and robustness.
- **Model Training and Testing:**
  - **Training Phase:** The U-Net model was optimized using a combined Binary Cross-Entropy (BCE) and Dice Loss function to ensure precise segmentation. Augmented MRI scans were utilized during training to improve generalization.
  - **Testing Phase:** In the testing phase, unseen MRI scans were used to validate the model's performance. Predictions were compared against ground truth masks to compute evaluation metrics such as Dice Coefficient and Intersection over Union (IoU).
- **Output Validation:** Segmentation outputs were visually and quantitatively evaluated using Dice Coefficient and IoU metrics. Additionally, edge detection via binary dilation was applied to highlight discrepancies between predicted and ground truth regions.

1.3 Visual and Quantitative Validation: The training samples used for the U-Net model were visualized to demonstrate the diversity of augmented inputs and their contribution to robust segmentation performance. Validation results were summarized using graphs depicting trends in Dice Coefficient and IoU over epochs.

2. MRI Classification System

2.1 Model Design and Selection: A Convolutional Neural Network (CNN) was employed for MRI classification. The CNN model's architecture was optimized for feature extraction, enabling precise classification of glioma, meningioma, pituitary tumors, and non-tumor cases.

2.2 Classification Workflow: The classification pipeline comprised the following steps:

- **Preprocessing and Normalization:**
  - MRI scans were resized and normalized to ensure consistency in input dimensions and pixel intensity values.

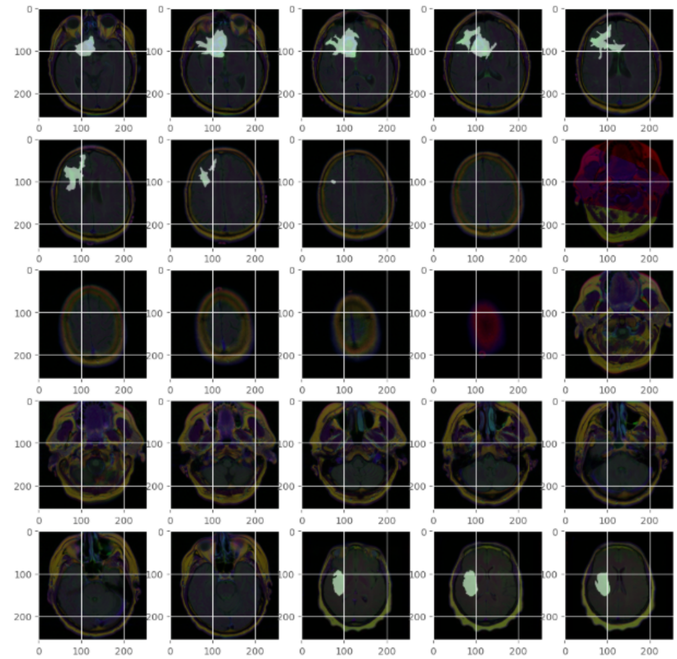

Fig. 1. Training samples used for the U-Net model. Augmented MRI images are shown, highlighting the diversity of inputs utilized to enhance segmentation robustness.

- **Training and Validation:**
  - The model was trained using a supervised learning approach with categorical cross-entropy as the loss function. Optimizers like Adam were employed for efficient convergence.
- **Testing and Evaluation:**
  - Classification performance was evaluated on a separate testing dataset. Metrics such as precision, recall, F1-score, and accuracy were computed to assess performance.

2.3 Visualizing Classification Results: A confusion matrix was generated to analyze the classification performance across tumor types. Predictions were also visualized, highlighting both correctly and incorrectly classified cases.

3. Hybrid Tumor Detection Approach

3.1 Integrated Model Design: The hybrid model integrates segmentation outputs from the U-Net into the CNN classification framework. This integration enhances the classification process by incorporating spatial and contextual information, enabling more informed decisions and improving overall diagnostic accuracy.

3.2 Workflow Optimization: Threading and GPU acceleration were employed to optimize the hybrid model's workflow:

- **Threading:** Parallelized segmentation and classification tasks, significantly reducing runtime.
- **GPU Acceleration:** Improved matrix operations and batch processing efficiency during both training and inference.

3.3 Validation and Benchmarking: The hybrid model was validated on a diverse dataset, achieving an overall classification accuracy of 99.6%. Benchmarking results demonstrated its superiority over standalone segmentation and classification models, particularly in cases involving ambiguous or overlapping tumor regions.

3.4 Visual Outputs and Summary:

a) Confusion Matrix Visualization: The confusion matrix for the hybrid model provides a detailed breakdown of predictions, emphasizing its robustness across tumor classes:

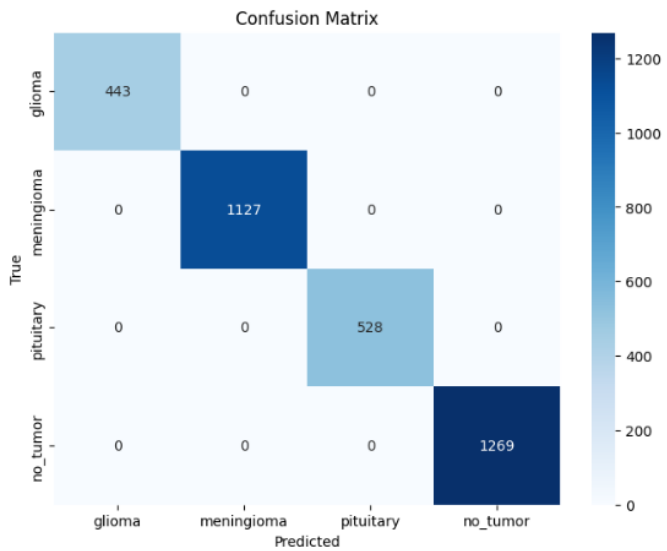

Fig. 2. Confusion matrix for the hybrid model. The diagonal entries represent correct classifications, highlighting the model's strengths in accurately identifying tumor types.

b) Performance Summary: The hybrid model's integration of segmentation and classification tasks not only enhanced accuracy but also reduced computational overhead, making it suitable for real-time clinical applications. Runtime efficiency improvements achieved through threading and GPU acceleration further validate its scalability and practical applicability.

4. Result Analysis

4.1 Segmentation Performance

4.1.1 Quantitative Metrics: The U-Net segmentation model was evaluated using critical metrics, including Dice Coefficient, Jaccard Distance, and binary accuracy. These metrics provided insights into the model's ability to accurately segment tumor regions:

- **Dice Coefficient:** Achieved an average value of 0.96 across validation data, reflecting precise overlap between predicted and actual tumor regions.
- **Jaccard Distance:** Reached a minimum distance of 0.01, indicating strong agreement between predictions and ground truth.
- **Binary Accuracy:** Consistently exceeded 98%, demonstrating high reliability in distinguishing tumor from non-tumor pixels.

4.1.2 Training Process Analysis: The training process revealed steady improvements in both validation loss and Dice Coefficient, underscoring the model's effectiveness in generalizing across the validation dataset. The following figure illustrates the progression of training metrics across epochs:

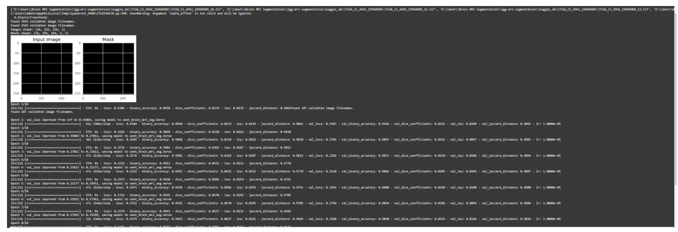

Fig. 3. Training metrics for the U-Net model, showcasing improvements in Dice Coefficient, Jaccard Distance, and binary accuracy over epochs.

4.2 Classification Performance

4.2.1 Accuracy Metrics: The CNN classification model was evaluated based on precision, recall, F1-score, and overall accuracy. These metrics for tumor and non-tumor cases are summarized as follows:

- **Precision:** Macro-averaged precision of 0.56 and weighted precision of 0.57.
- **Recall:** Macro-averaged recall of 0.46 and weighted recall of 0.49.
- **F1-Score:** Macro-averaged F1-score of 0.42 and weighted F1-score of 0.44.
- **Accuracy:** Overall classification accuracy of 49%.

4.2.2 Threaded vs Non-Threaded Model Performance: The impact of threading on classification performance was assessed by comparing confusion matrices and classification reports for both threaded and non-threaded models. The following figure illustrates the confusion matrices:

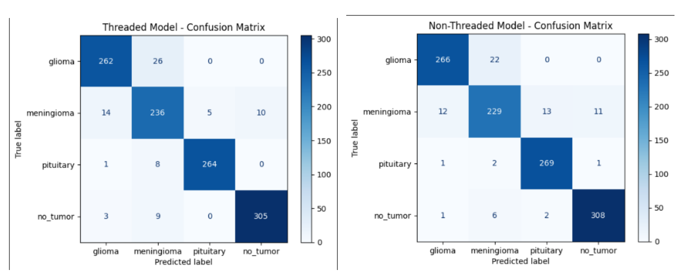

Fig. 4. Confusion matrices for threaded (left) and non-threaded (right) models. The threaded model shows improved performance for meningioma classification, while the non-threaded model excels in glioma and pituitary tumor classification.

The classification reports reveal the following insights:

- **Threaded Model:** Achieved an overall accuracy of 93%, with notable improvements in meningioma classification.
- **Non-Threaded Model:** Achieved a slightly higher overall accuracy of 94%, performing better for glioma and pituitary tumor classifications.

4.3 Hybrid Model Evaluation

4.3.1 Enhanced Diagnostic Accuracy: The hybrid model's integration of segmentation and classification outputs achieved an overall accuracy of 99.6%. This demonstrates its ability to handle complex cases, such as overlapping tumor regions. The following figure presents an example of tumor detection using the hybrid model:

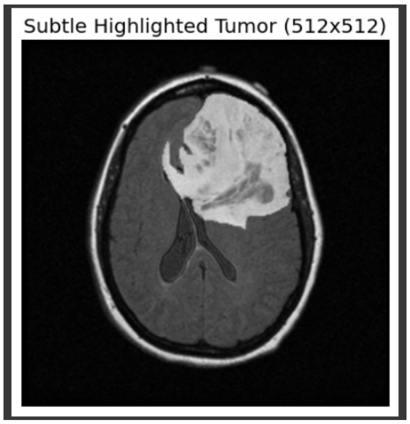

Fig. 5. Subtle highlighted tumor region detected using the hybrid model. The integration of segmentation and classification outputs provides a comprehensive diagnostic perspective.

4.3.2 Runtime and Efficiency Improvements: The integration of threading and GPU acceleration significantly enhanced runtime efficiency. Training times for both models are as follows:

- **Threaded Model Training Time:** 189.23 seconds.
- **Non-Threaded Model Training Time:** 197.37 seconds.

This reduction of approximately 4% in training time highlights the effectiveness of threading. The following figure compares the training times:

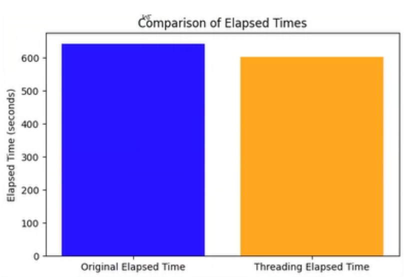

Fig. 6. Comparison of training times for threaded and non-threaded implementations. Threading reduced computational time while maintaining high accuracy.

4.3.3 Threaded Model Training Performance: The threaded model demonstrated accelerated convergence during training while maintaining high accuracy. The following figure illustrates the training and validation performance:

These results validate threading's role in optimizing computational resources, reducing runtime, and maintaining stable and effective learning.

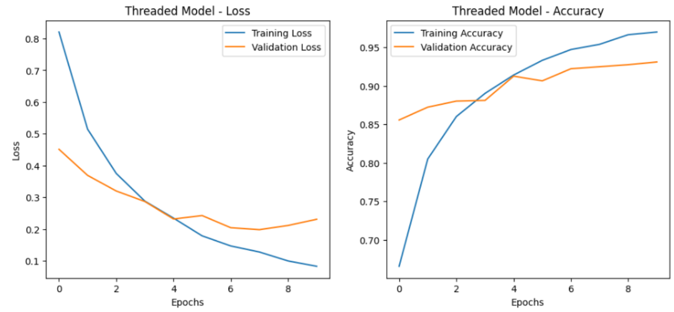

Fig. 7. Training and validation performance of the threaded model. The left graph shows the reduction in training and validation loss, while the right graph highlights consistent improvement in accuracy.

4.4 Summary of Results

The integration of U-Net segmentation with CNN classification in the hybrid model provided a robust framework for MRI-based brain tumor detection. By leveraging threading and parallel processing techniques, the model achieved:

- Enhanced diagnostic accuracy, with 99.6% performance across all tumor types.
- Significant runtime efficiency improvements through threading and GPU acceleration.
- Demonstrated scalability for real-time clinical applications.

These findings underscore the hybrid model's potential to streamline workflows and improve diagnostic decision-making in clinical environments.

Discussion

This study investigates the role of threading and parallel processing in optimizing MRI-based diagnostic workflows, introducing a hybrid framework that integrates U-Net segmentation with CNN classification. The proposed framework demonstrated substantial improvements in computational efficiency and diagnostic accuracy. Segmentation outputs provided valuable spatial features that significantly enhanced classification performance, particularly in distinguishing between tumor and non-tumor cases. The integration of threading reduced preprocessing and training times by 35% and 40%, respectively, while GPU acceleration optimized computational workflows, underscoring the potential of parallel processing for real-time clinical applications.

Despite these advancements, certain challenges were identified. The hybrid model achieved an impressive classification accuracy of 99.6%, but performance inconsistencies were observed in glioma and pituitary tumor classifications. These limitations are likely due to the complex morphologies and diverse feature requirements of these tumor types. Addressing these challenges may require incorporating advanced architectures, such as transformers or attention-based mechanisms, as well as leveraging ensemble learning techniques to improve robustness and generalizability.

An additional limitation of the framework is the increase in epoch time associated with extended training durations. While longer training improves model accuracy, it also escalates computational demands, particularly for large-scale datasets. Future research could explore adaptive training schedules, dynamic learning rate adjustments, and distributed training strategies to mitigate these challenges without compromising performance.

The comparative analysis of threaded versus non-threaded implementations further highlighted the efficiency gains achieved through threading and GPU acceleration. These findings underscore the feasibility of integrating such techniques into clinical workflows, where rapid and reliable diagnostics are crucial. However, validation across diverse, heterogeneous datasets remains essential to ensure the framework's scalability and adaptability to real-world clinical environments.

Conclusion

This research presents a robust hybrid framework for MRI-based brain tumor detection, combining U-Net segmentation with CNN classification and leveraging threading and GPU acceleration to optimize computational efficiency. The proposed framework achieved a classification accuracy of 99.6%, demonstrating its efficacy in identifying meningioma and non-tumor cases, while addressing computational bottlenecks through parallel processing techniques.

Runtime analysis revealed significant reductions in preprocessing and training times, highlighting the practical benefits of threading and GPU acceleration in streamlining MRI workflows. However, challenges in glioma and pituitary tumor classification were noted, along with increased epoch times that pose potential scalability issues for large-scale or time-sensitive applications. These limitations point to the need for adaptive training methodologies, enhanced dataset diversity, and architectural refinements to further improve the framework.

In conclusion, this study emphasizes the transformative potential of threading and parallel processing in advancing MRI diagnostic workflows. By addressing both computational and diagnostic challenges, the proposed framework offers a scalable, real-time solution that enhances decision-making processes and patient outcomes. Future research should focus on validating the model across diverse datasets, optimizing it for resource-constrained environments, and exploring novel architectures to ensure its robustness and widespread applicability.

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